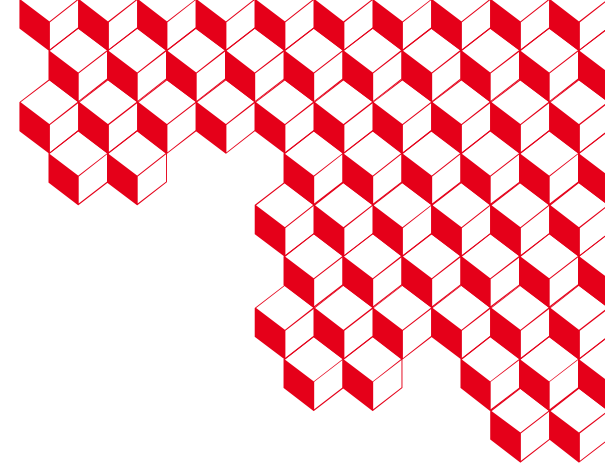




Towards Baseline-Free Guided-Wave Damage Characterization Using Multimodal Deep Learning

Amira Yassine – Robin Guyon – Guillemette Ribay



list

French Alternative Energies and Atomic Energy Commission.

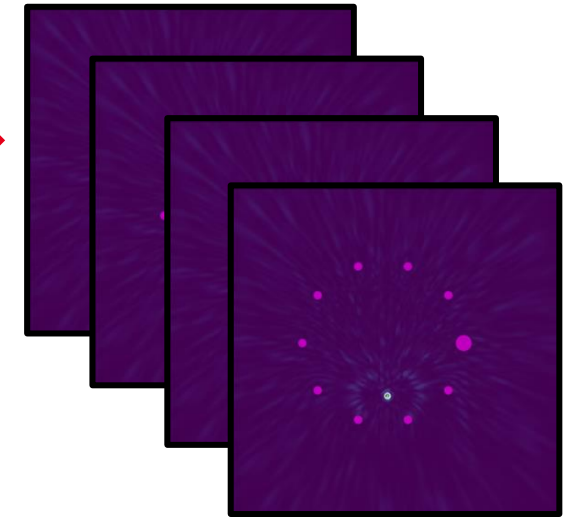
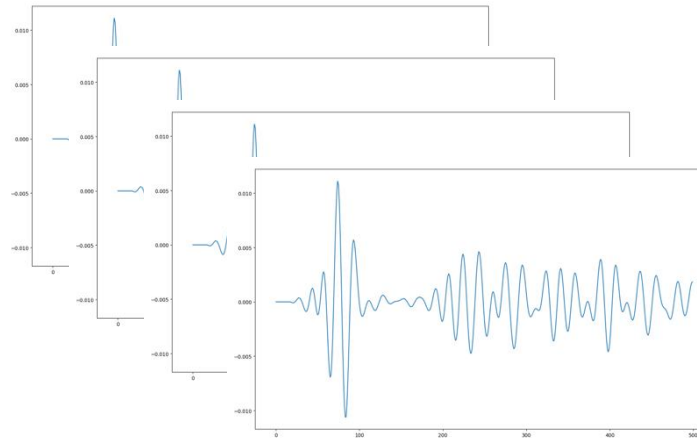
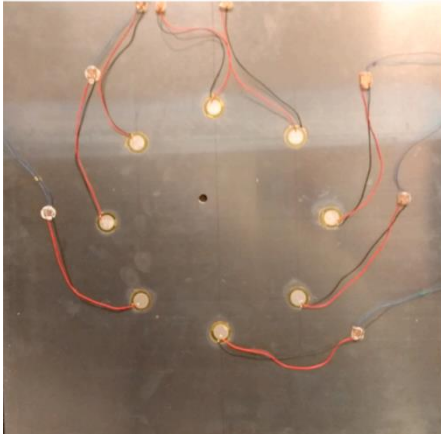
Laboratoire d'Intégration des Systèmes et des Technologies



1. A transmitter emits a guided wave through the plate.
2. The other sensors receive the wave.

• The signals alone do not allow the defect to be visually identified, so they are combined to generate a more interpretable image.

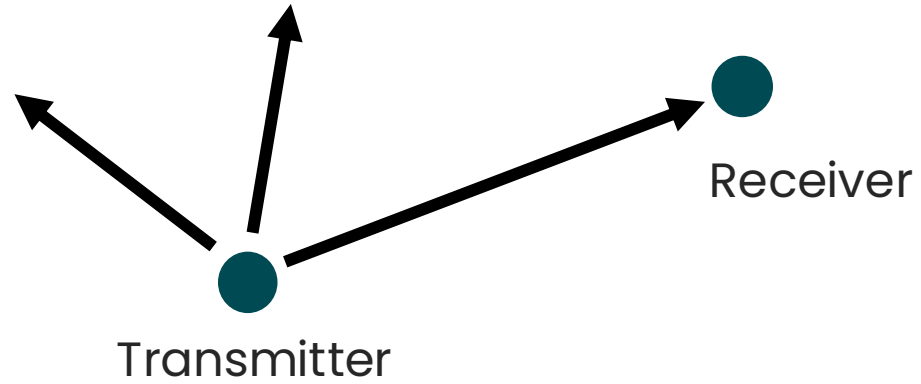
1. Images are mainly useful for defect localization.
2. Limited characterization: defect type and depth are difficult to identify.
3. Limited spatial resolution: accuracy is typically on the order of $\lambda/2$ at best.



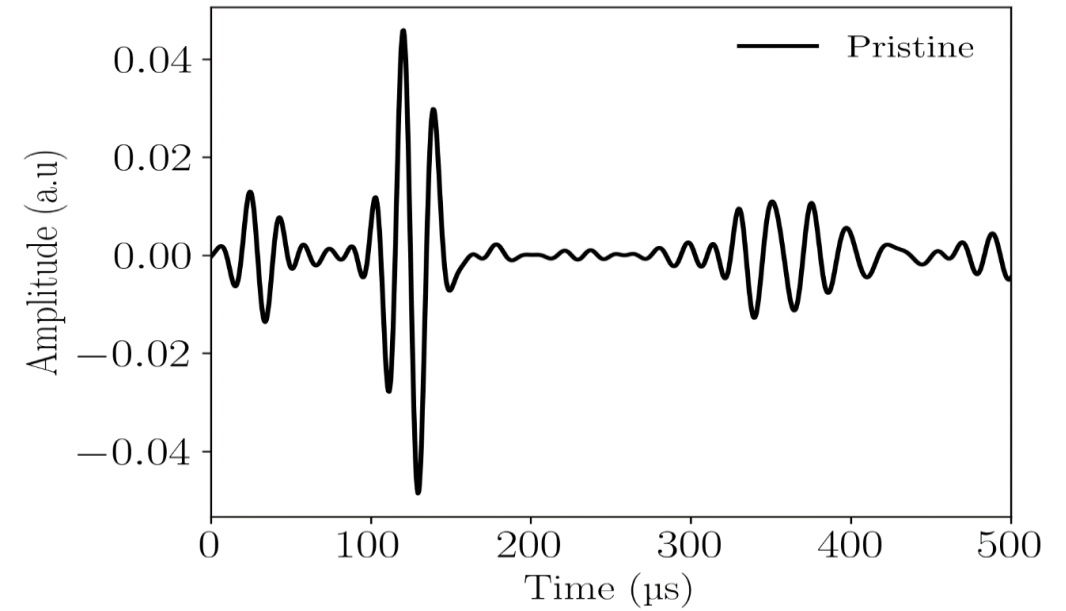
Goislot, P. (2026). *Apport des méthodes vibro-acoustiques non linéaires à la détection et à la localisation d'endommagements dans les structures aéronautiques en matériaux composites*. PhD thesis, Université Polytechnique Hauts-de-France.

Acoustic signature of a defect

Baseline state

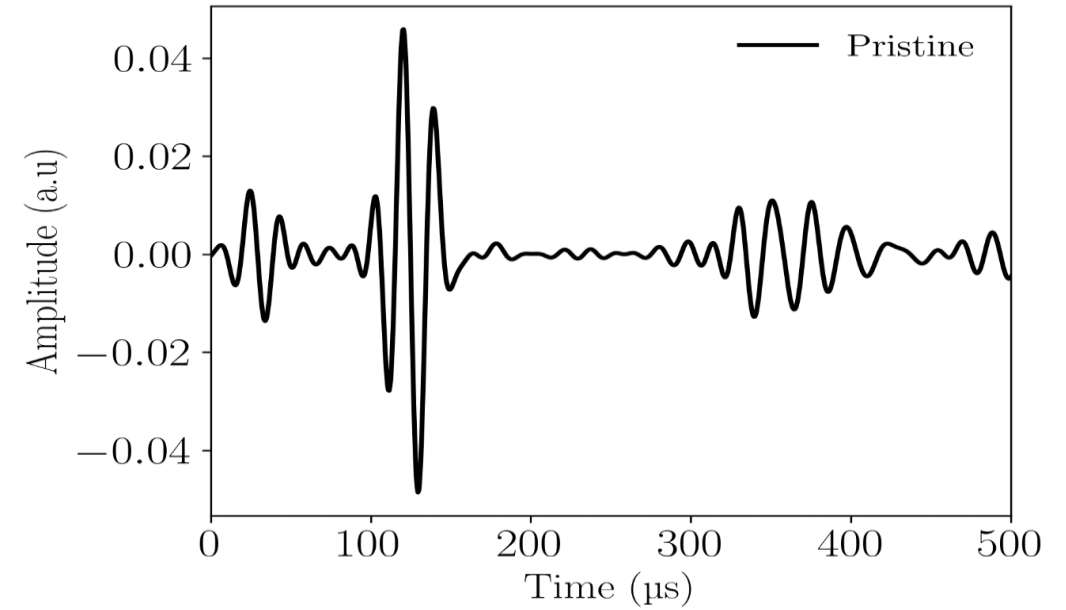
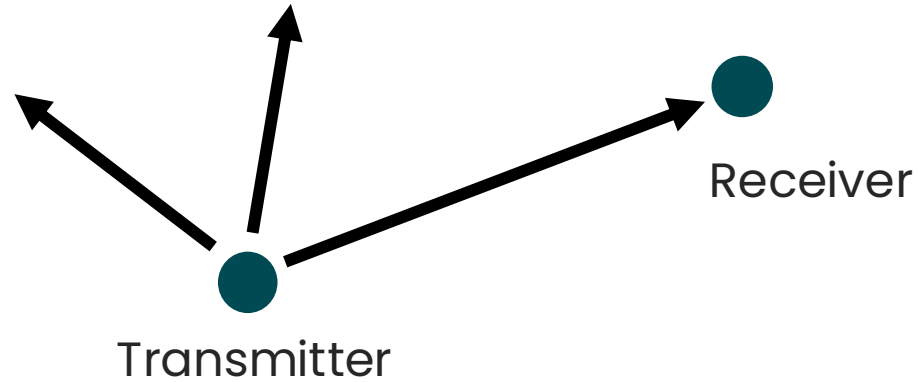


Damaged state

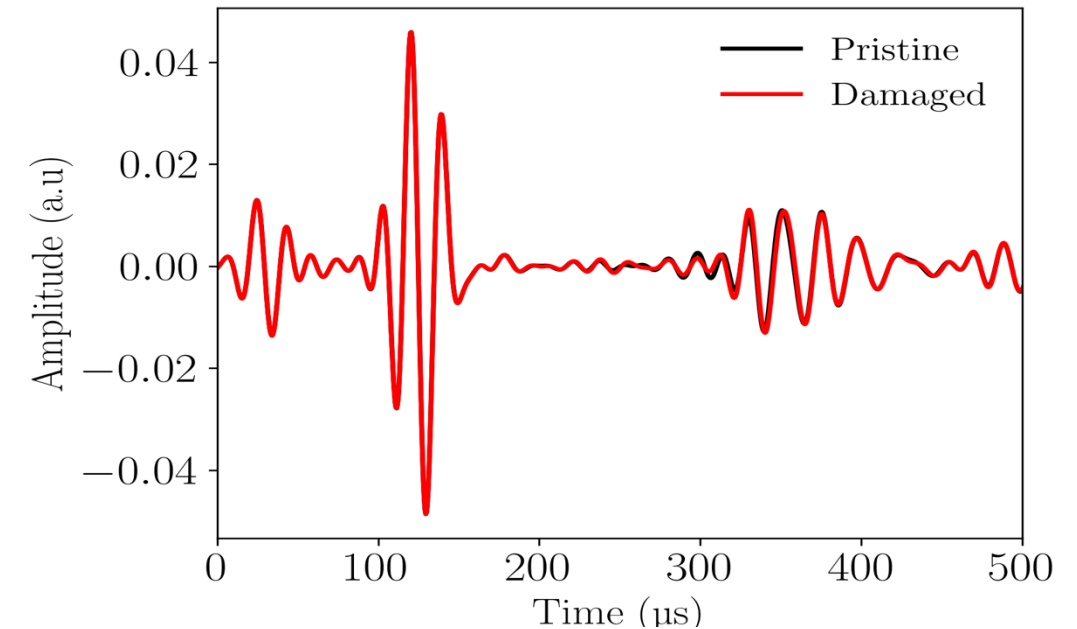
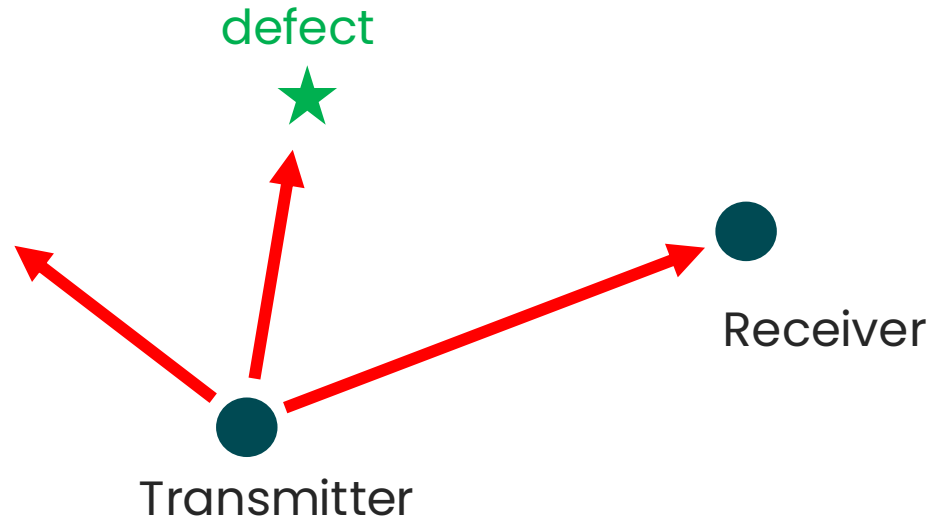


Acoustic signature of a defect

Baseline state

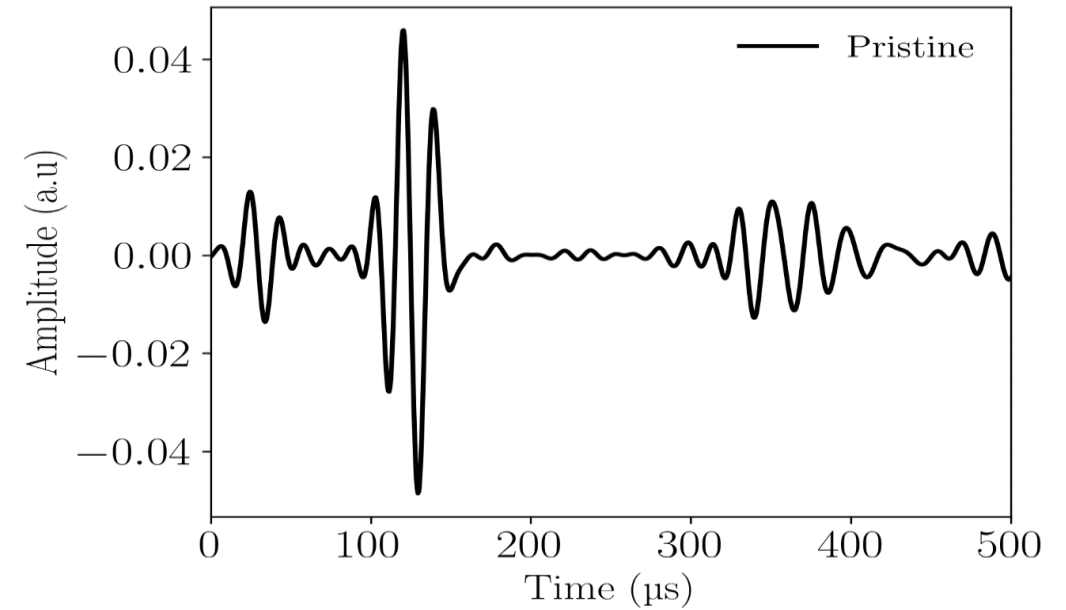
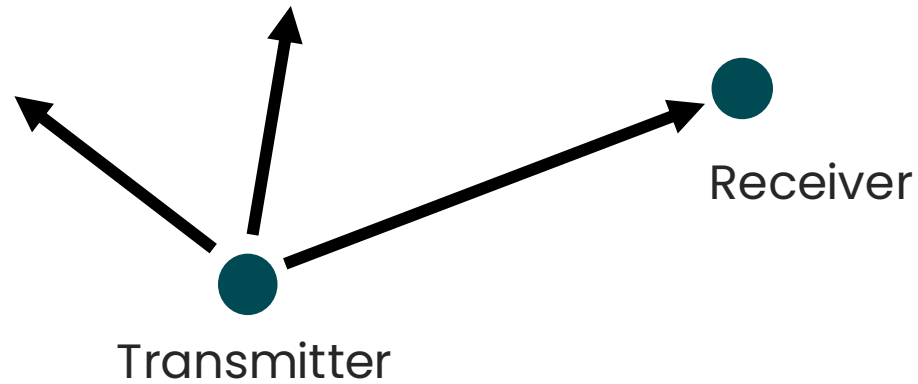


Damaged state

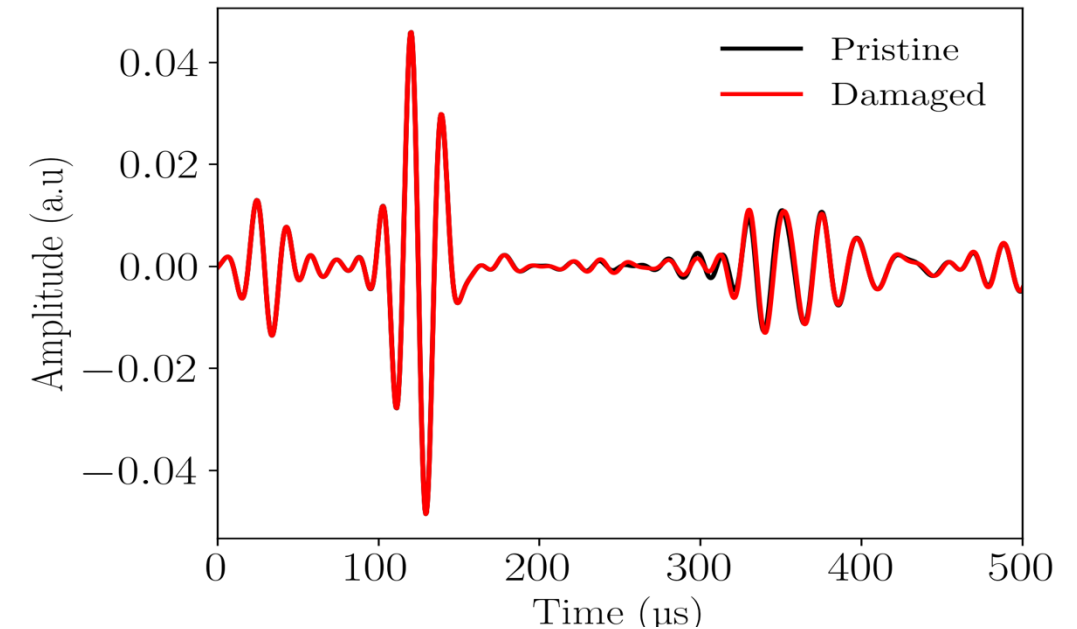
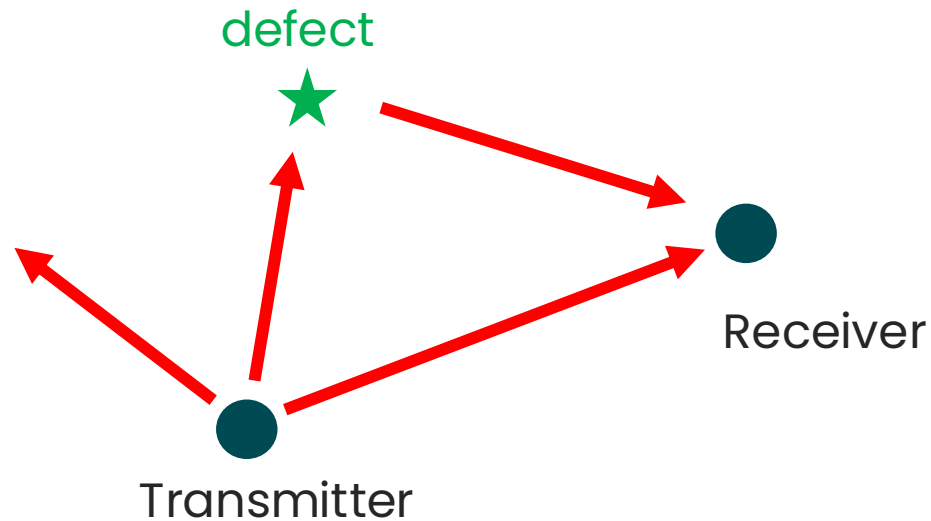


Acoustic signature of a defect

Baseline state

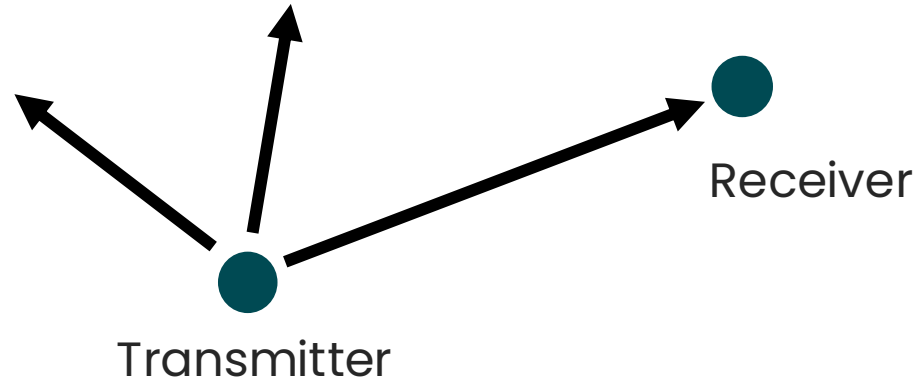


Damaged state

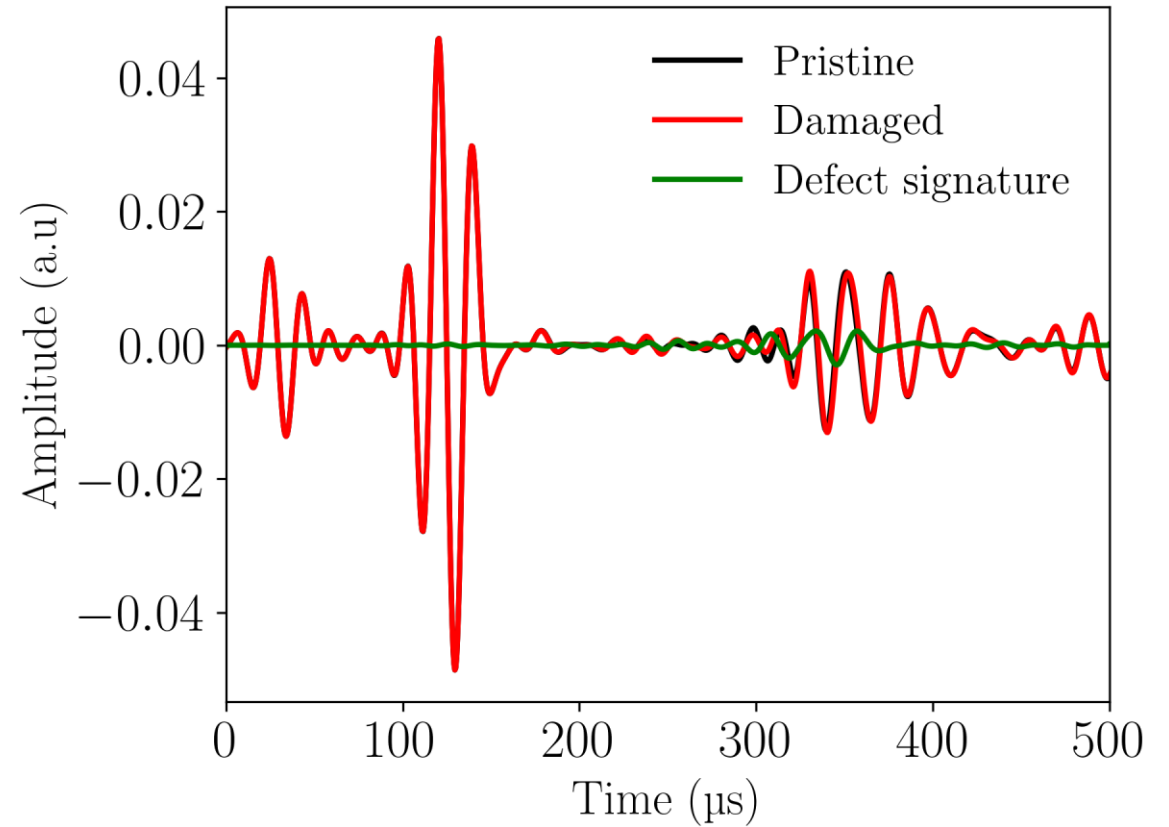
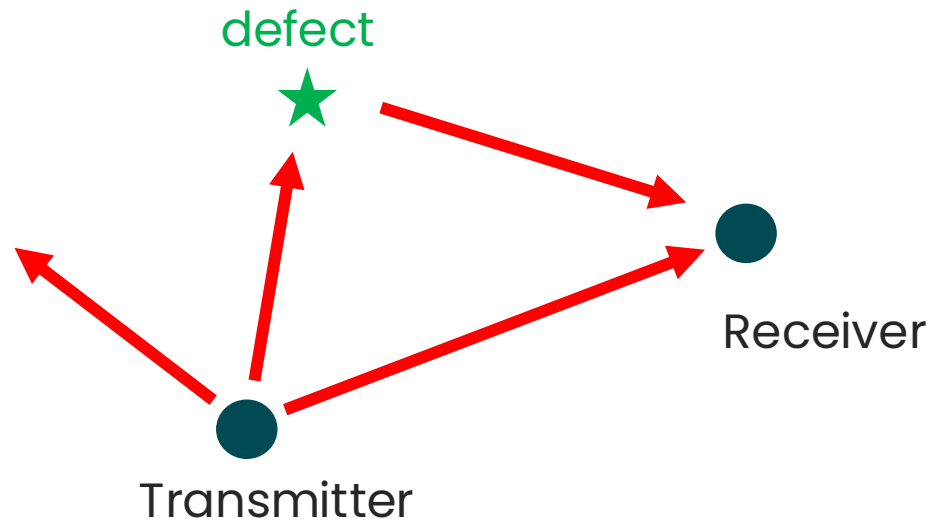


Acoustic signature of a defect

Baseline state



Damaged state



Our goals:

- Investigate the use of neural networks to localize and characterize defects in a plate.
- Use raw simulated signals generated with CIVA, together with defect maps.

Motivation:

1. Avoid the need for healthy reference signals.
2. Enable defect characterization, including defect classification and parameter estimation.

Two types of plates used for simulation:

- **Aluminum plate**
- Composite plate

Types of defects:

- **For the aluminum plate: Hole and simulated “delamination”**
- For the composite plate: Delamination

Plate and defect type	Number of configurations	Number of transmitter– receiver signals per configuration	Signal length	Parameters
Aluminum – Hole	2500	90	500	Radius, Y position, X position
Aluminum – Delamination	2496	90	500	Depth, Rx, Ry, Y position, X position

Characteristics of the aluminum plate:

- $E = 69.98 \text{ Gpa}$
- $\nu = 0.344$
- $\rho = 2.71 \text{ g/cm}^3$
- thickness = 2.54 mm
- $L_x, L_y = 609 \text{ mm}$
- freq=50 kHz
- 3 cycles
- $\lambda = 21 \text{ mm}$
- 10 piezoelectric sensors



The data must be preprocessed before being used in AI algorithms.

Outline :

1. Data analysis and preprocessing

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- a) Results for hole-type defects
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3. Deep learning: supervised models using defect images

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4. Multi-Head Autoencoder

5. Conclusion

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- **1 st step:** Conversion of .cck files into .hdf5 format
- **2 nd step:** Data normalization between 0 and 1

For time-series signals

$$x_{norm} = \frac{x - x_{min,config}}{x_{max,config} - x_{min,config}}$$

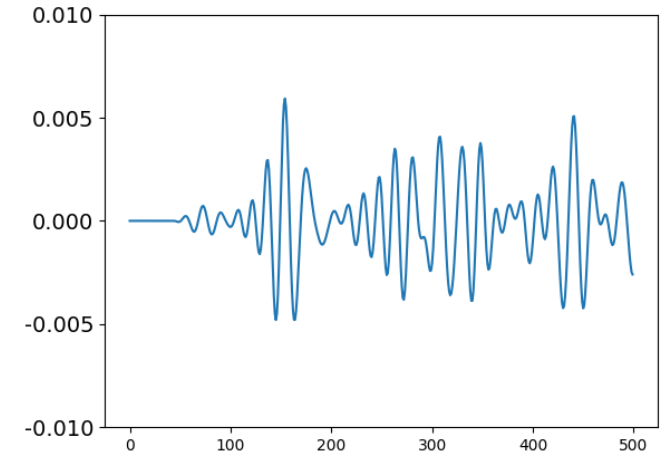
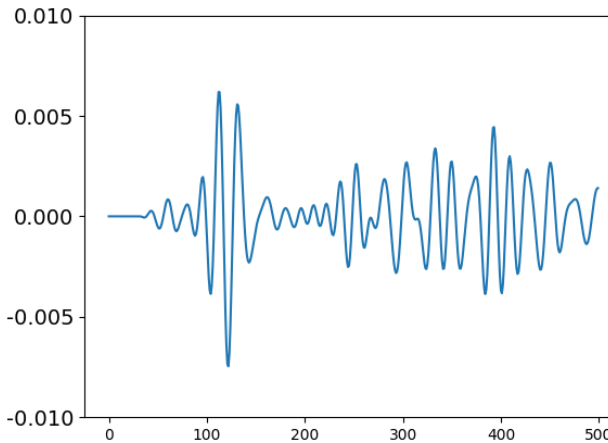
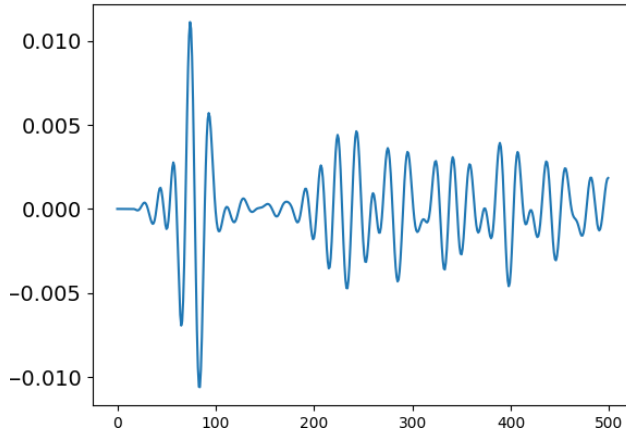
For image pixels

$$p_{norm} = \frac{p - p_{min,image}}{p_{max,image} - p_{min,image}}$$

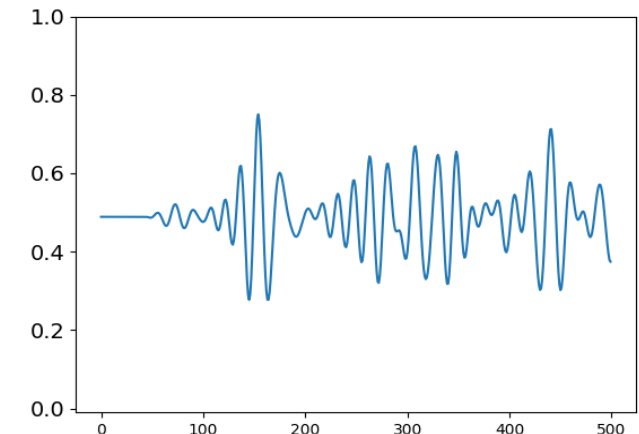
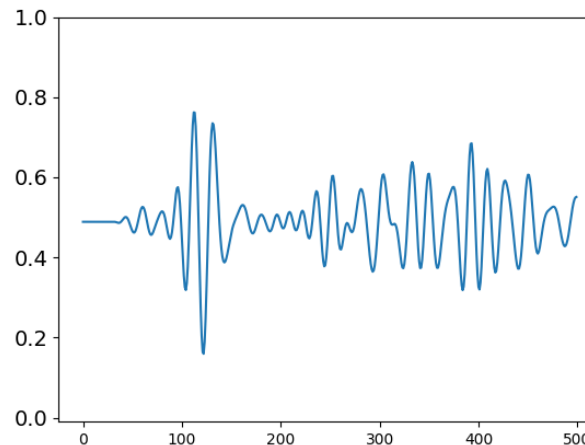
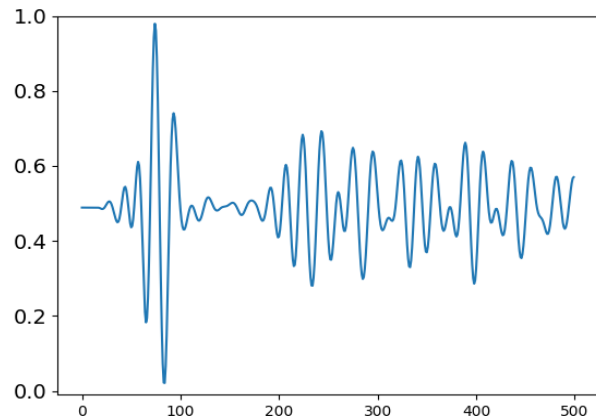
For defect parameters

$$y_{norm} = \frac{y - y_{min}}{y_{max} - y_{min}}$$

Before normalisation



After normalization





- **1 st step:** Conversion of .cck files into .hdf5 format
- **2 nd step:** Data normalization between 0 and 1

For time-series signals

$$x_{norm} = \frac{x - x_{min,config}}{x_{max,config} - x_{min,config}}$$

For image pixels

$$p_{norm} = \frac{p - p_{min,image}}{p_{max,image} - p_{min,image}}$$

For defect parameters

$$y_{norm} = \frac{y - y_{min}}{y_{max} - y_{min}}$$

Evaluation metrics:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

- **MAE (Mean Absolute Error)** : Average of the absolute differences between the predicted values and the ground-truth values.
- **MSE (Mean Squared Error)**: Average of the squared errors, which gives more weight to large errors.
- **RMSE (Root Mean Squared Error)**: Square root of the MSE, expressed in the same unit as the predicted variable.
- **R^2 (Coefficient of determination)**: Measures the proportion of the variance in the data explained by the model; the closer it is to 1, the better the prediction.

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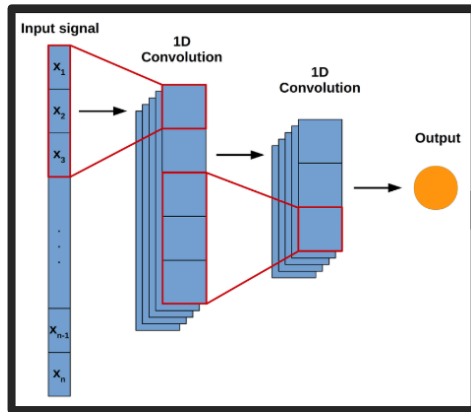
4. Multi-Head Autoencoder

5. Conclusion

Is it possible to eliminate the need for reference signals? (baseline state)

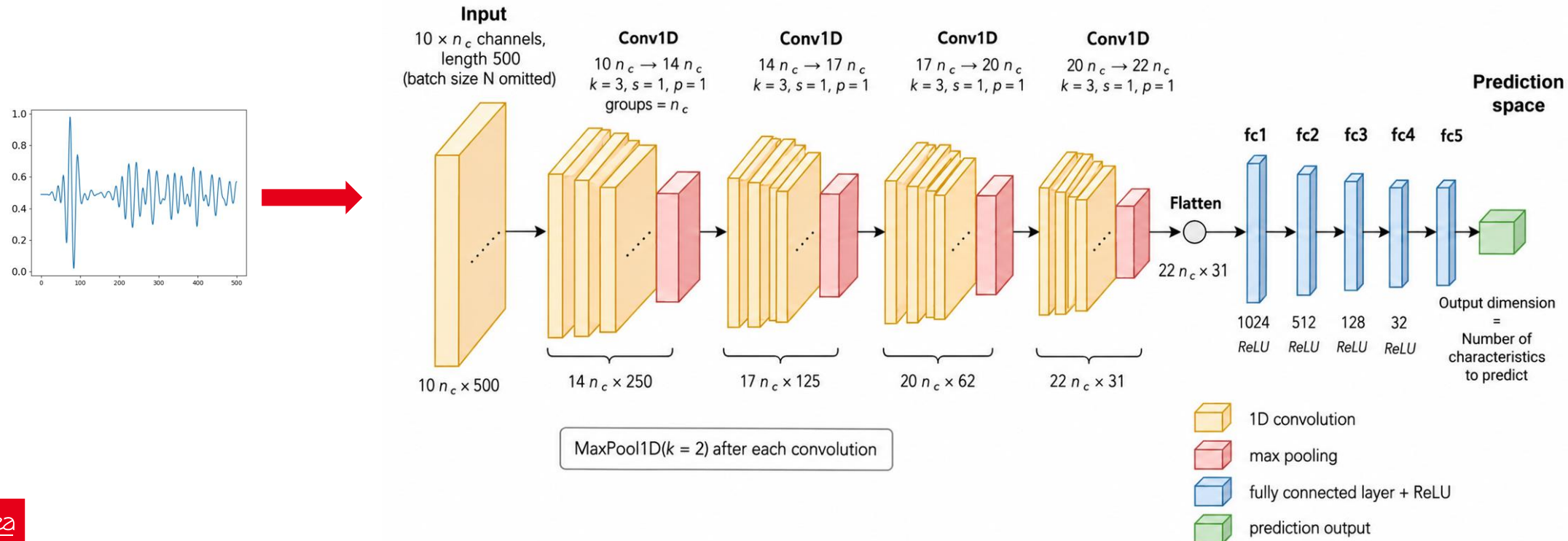
Models using time-domain signals in supervised learning

Progressive evolution toward increasingly complex architectures

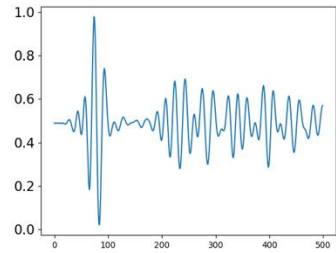


in_channels out_channels kernel_size stride padding

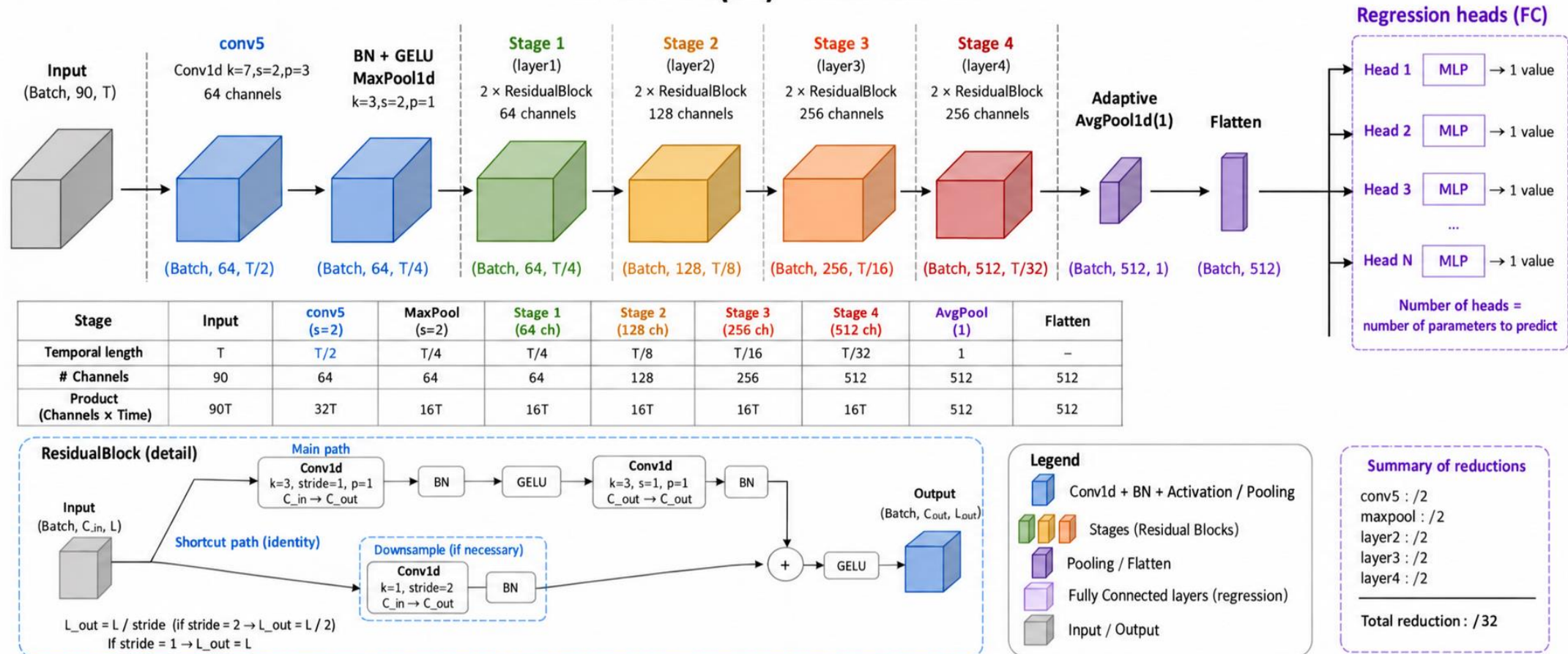
1D CNN architecture



ResNet18 1D



ResNet18 (1D) – Architecture

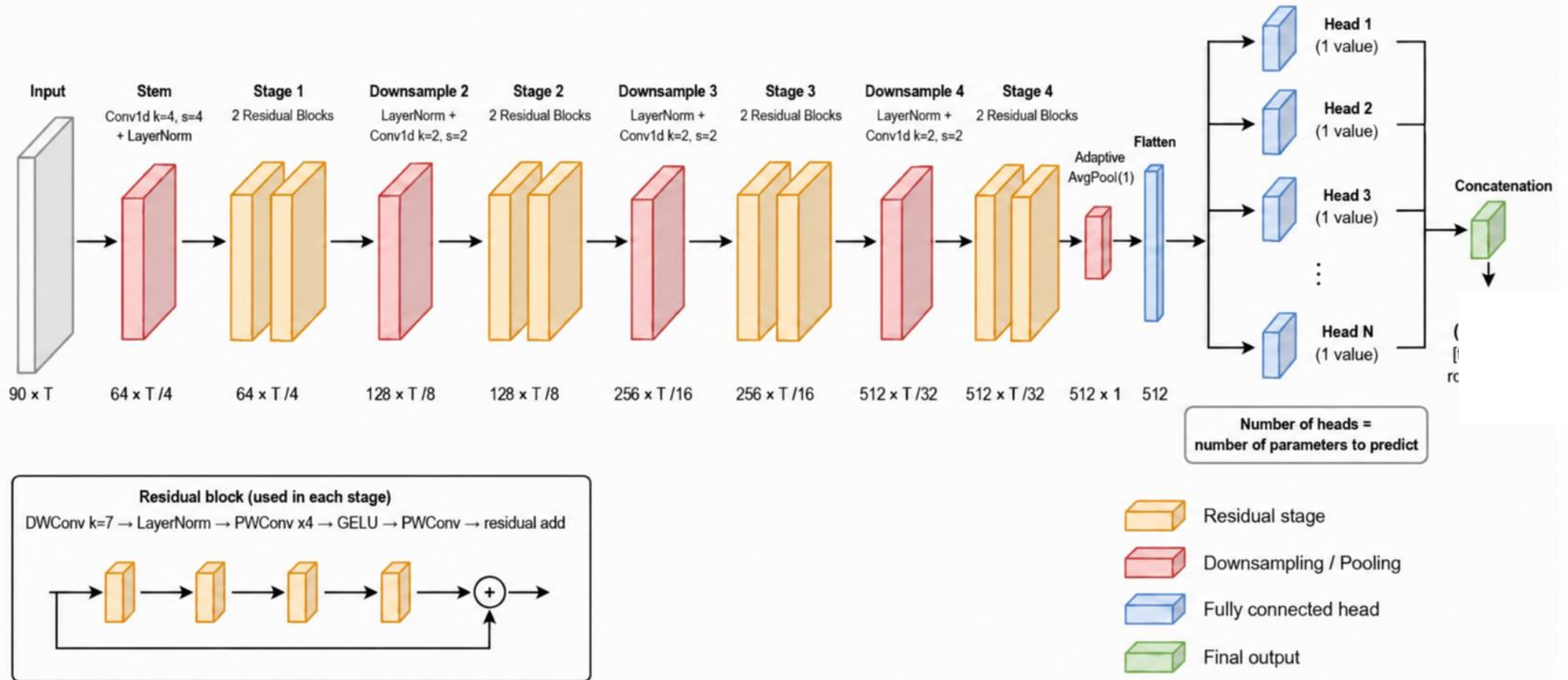
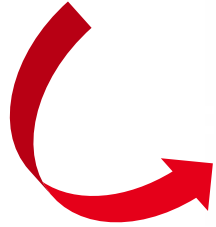
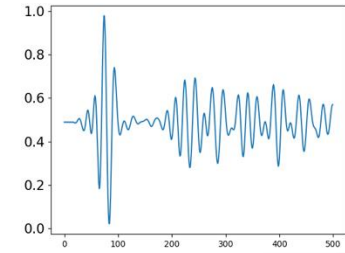


Source: He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep Residual Learning for Image Recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778.



AmiraNet

Fusion ResNet 18 1D – ConvNext 1D



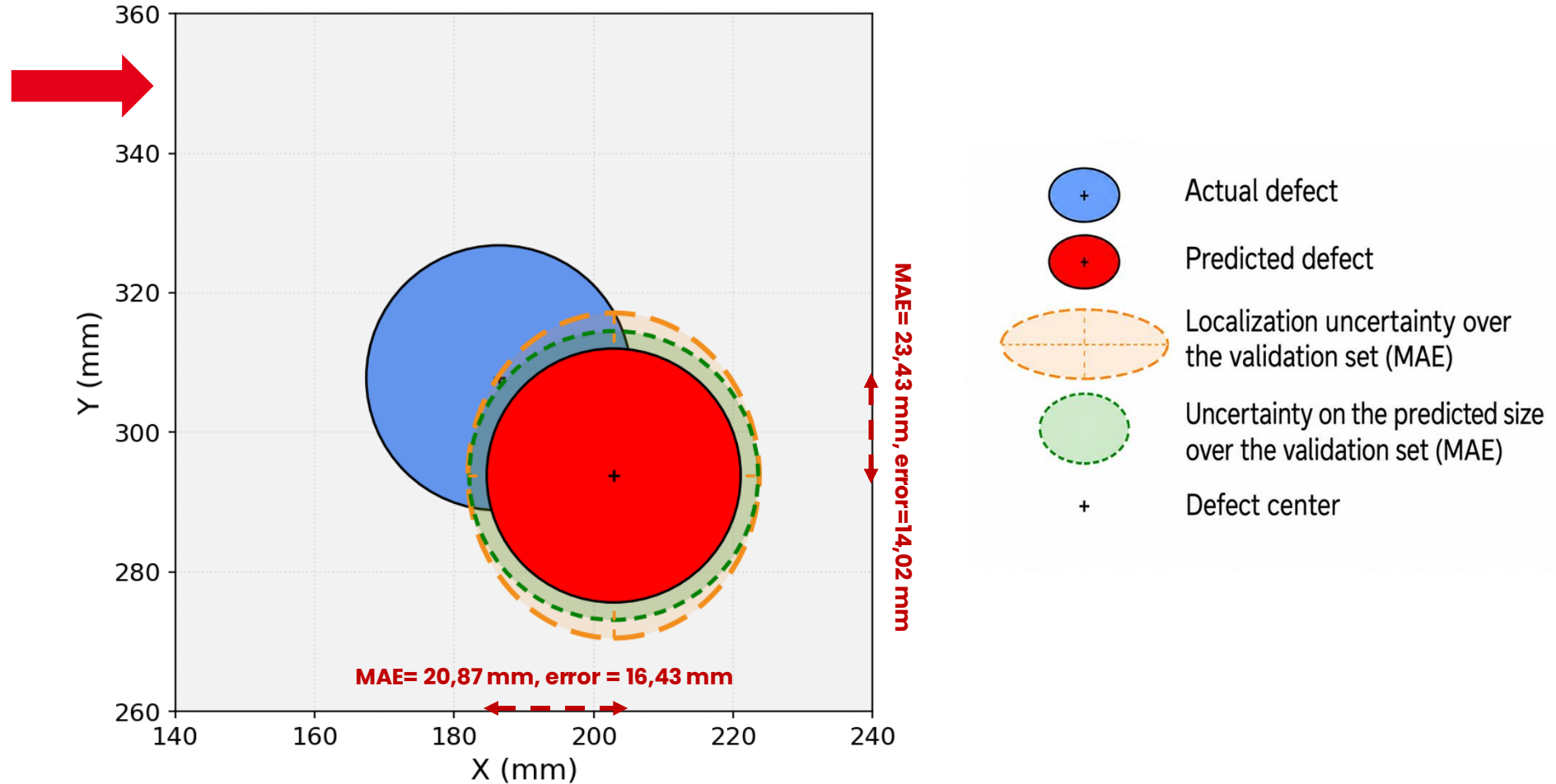
Source: Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A ConvNet for the 2020s. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 11966–11976.

ResNet18 1D - Hole

Metric	R^2		
Parameters	Hole radius	Pos Y	Pos X
Train	0.995	0.997	0.95
Val	0.78	0.85	0.75

Slight overfitting

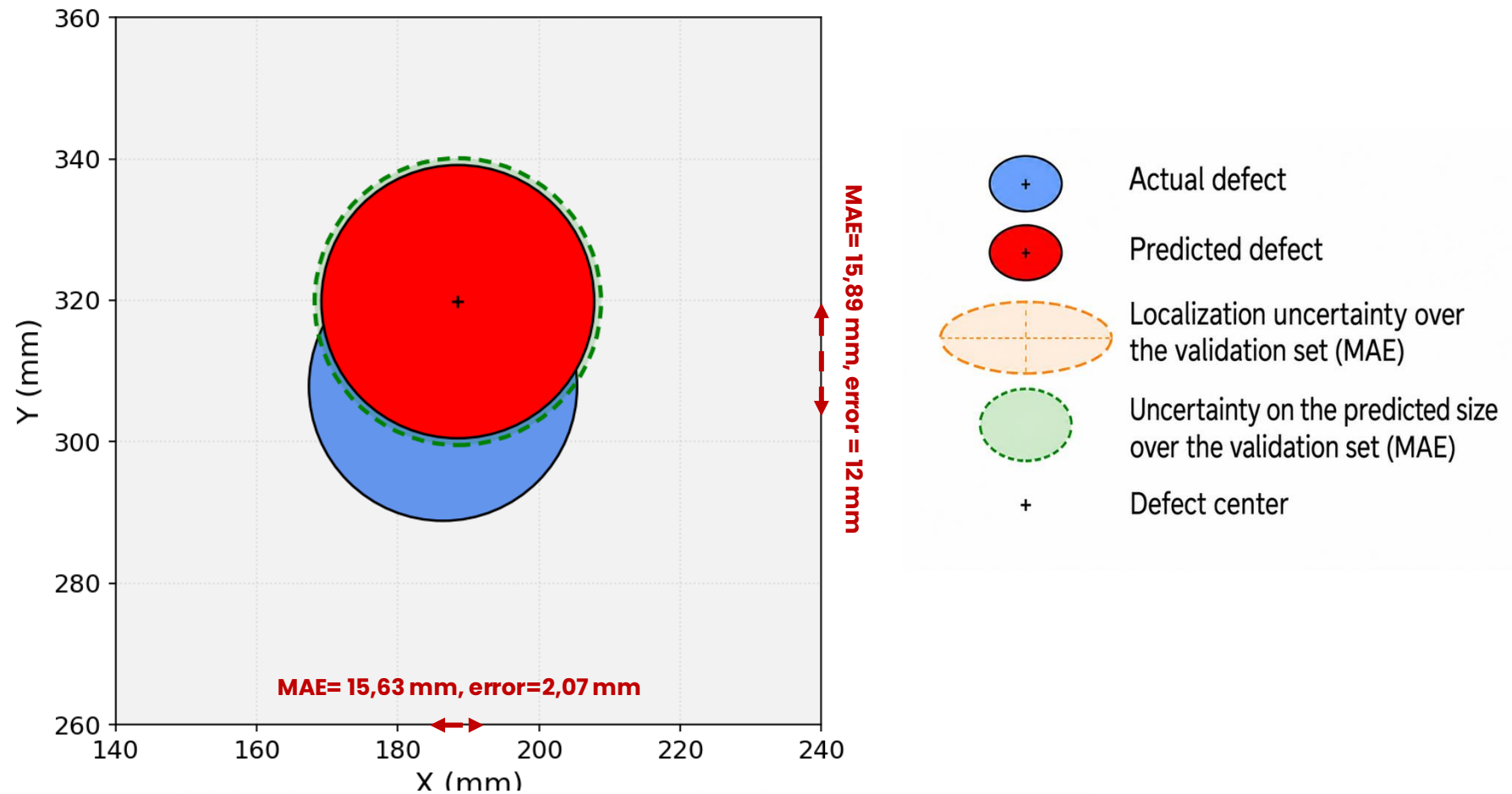
Actual defect from the validation set and the corresponding prediction obtained by the model.



Visualization of the actual defect, the predicted defect, and the associated uncertainties

AmiraNet - Hole

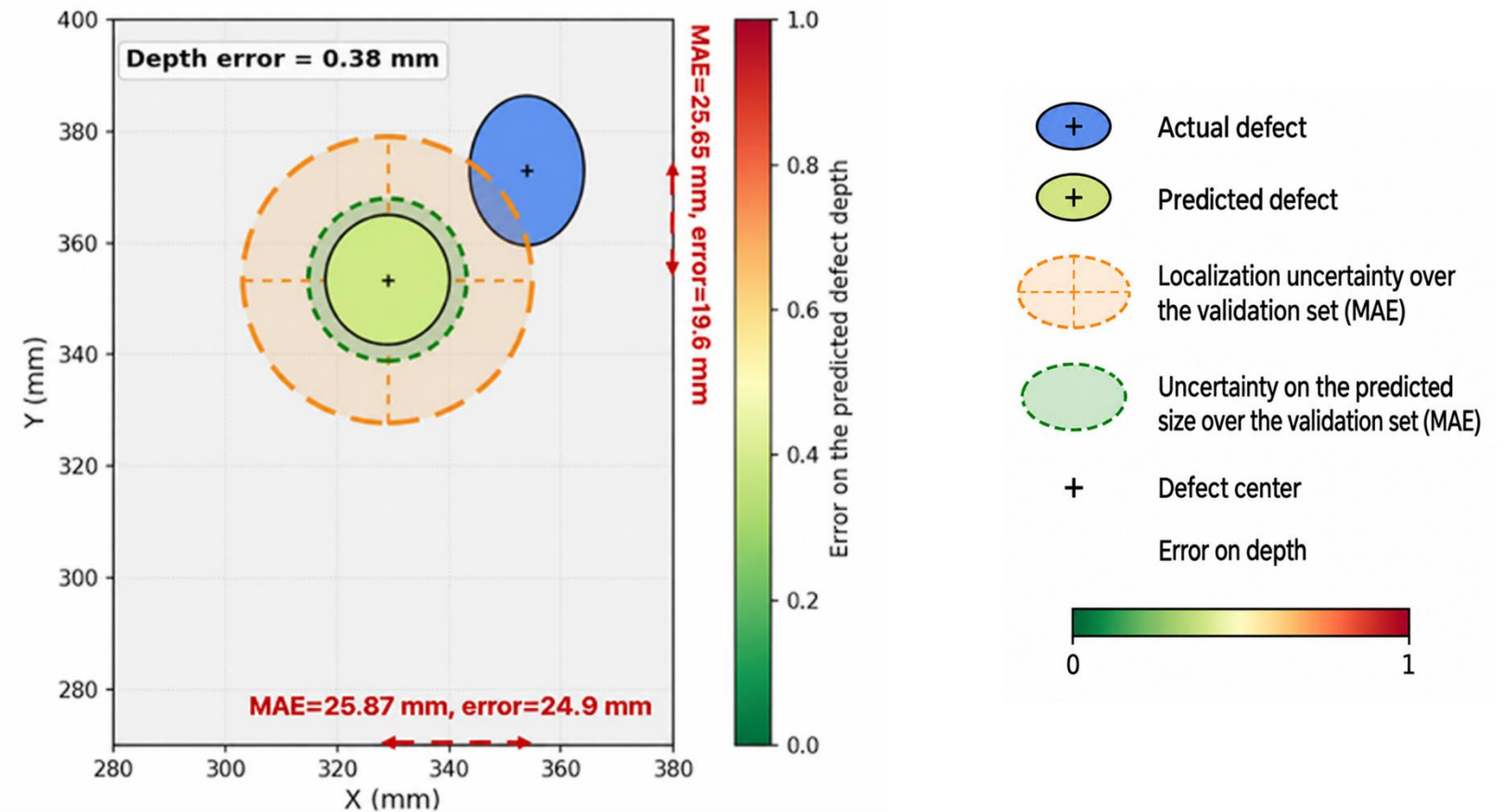
Metric	R^2		
Parameters	Rayon du Trou	Pos Y	Pos X
Train	0.94	0.97	0.95
Val	0.9	0.93	0.9



Visualization of the actual defect, the predicted defect, and the associated uncertainties

ResNet 18 1D - delamination

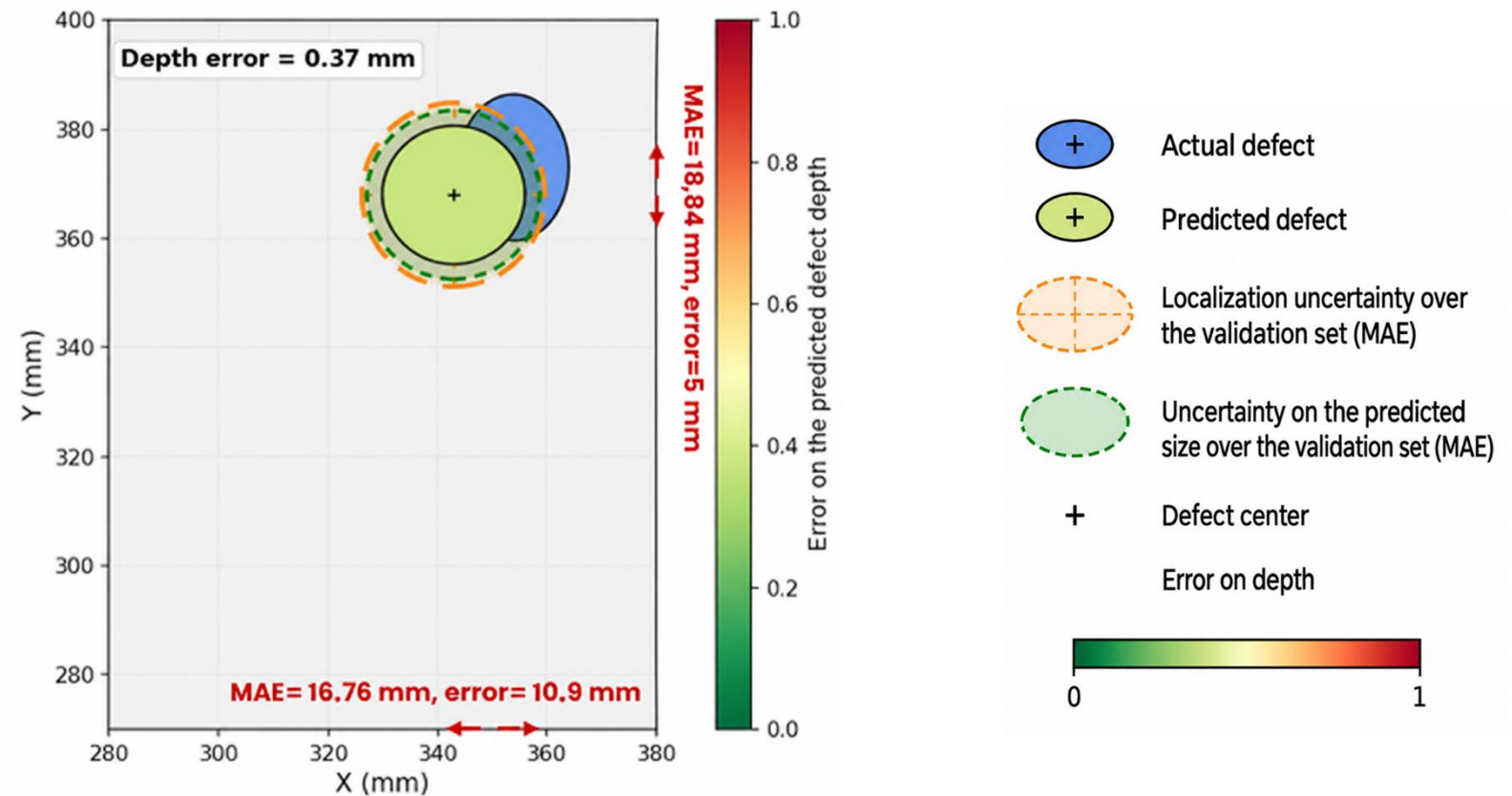
Metric	R^2				
Parameters	Depth	Rx	Ry	Pos Y	Pos X
Train	0.995	0.997	0.99	0.92	0.93
Val	-0.28	0.202	0.29	0.72	0.59



Visualization of the actual defect, the predicted defect, and the associated uncertainties

AmiraNet - delamination

Metric	R^2				
Parameters	Depth	Rx	Ry	Pos Y	Pos X
Train	0.07	0.45	0.45	0.91	0.9
Val	-0.02	0.39	0.41	0.86	0.83



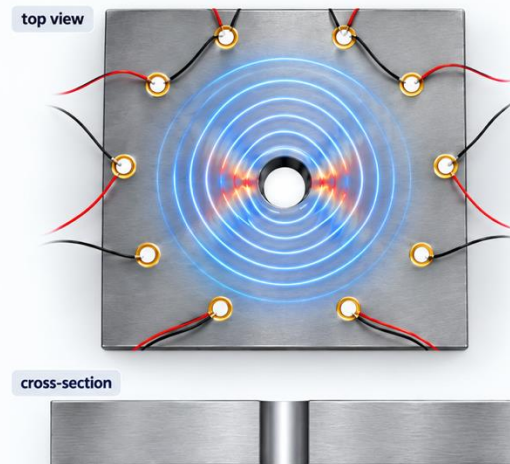
Observation:

- **For the hole defect:** Rather satisfactory results with AmiraNet.
- **For the delamination defect:**
 1. Depth is difficult to predict.
 2. Significant overfitting for the radii and the X and Y positions.

Conclusion: Eliminating the need for reference signals appears more feasible for the hole defect than for the delamination defect.

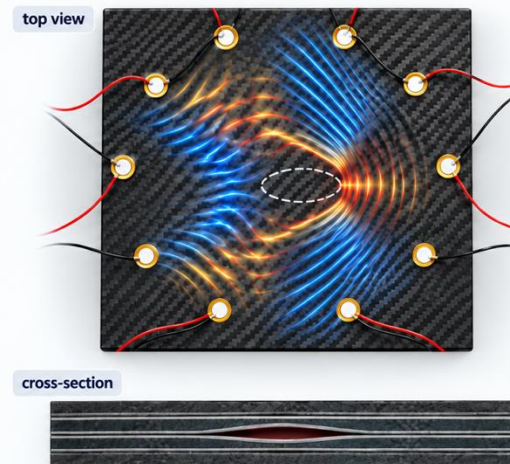
Why is a hole easier than a delamination?

Hole in aluminum



Hole: visible and simpler response

Delamination in composite



Delamination: hidden and more complex response

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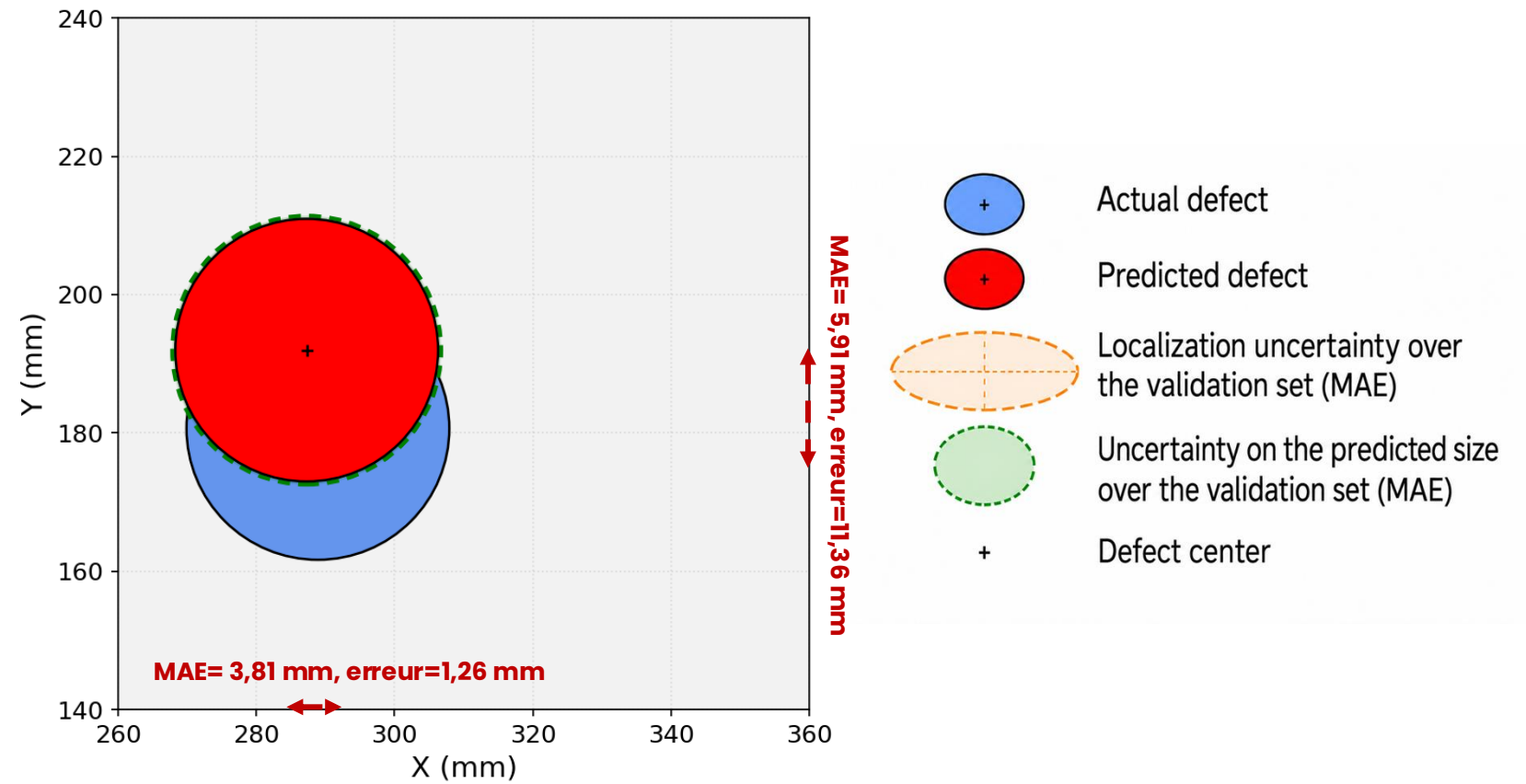
4. Multi-Head Autoencoder

5. Conclusion

To what extent can the use of a reference signal improve defect characterization?

ResNet 18 2D - Hole

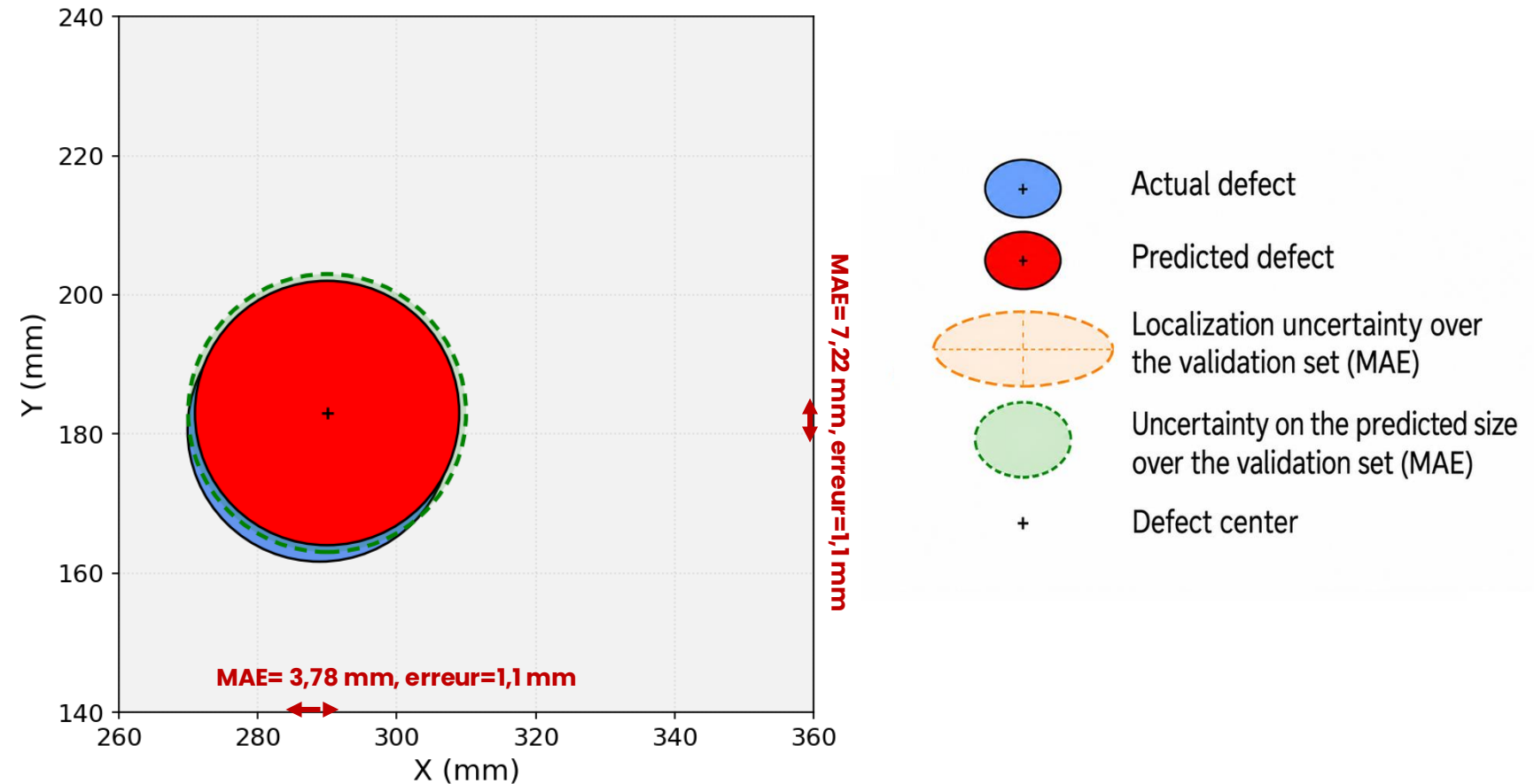
Metric	R^2		
Parameters	Hole Radius	Pos Y	Pos X
Train	0.73	0.928	0.99
Val	0.75	0.924	0.99



Visualization of the actual defect, the predicted defect, and the associated uncertainties

MobileNet V3 Small- Hole

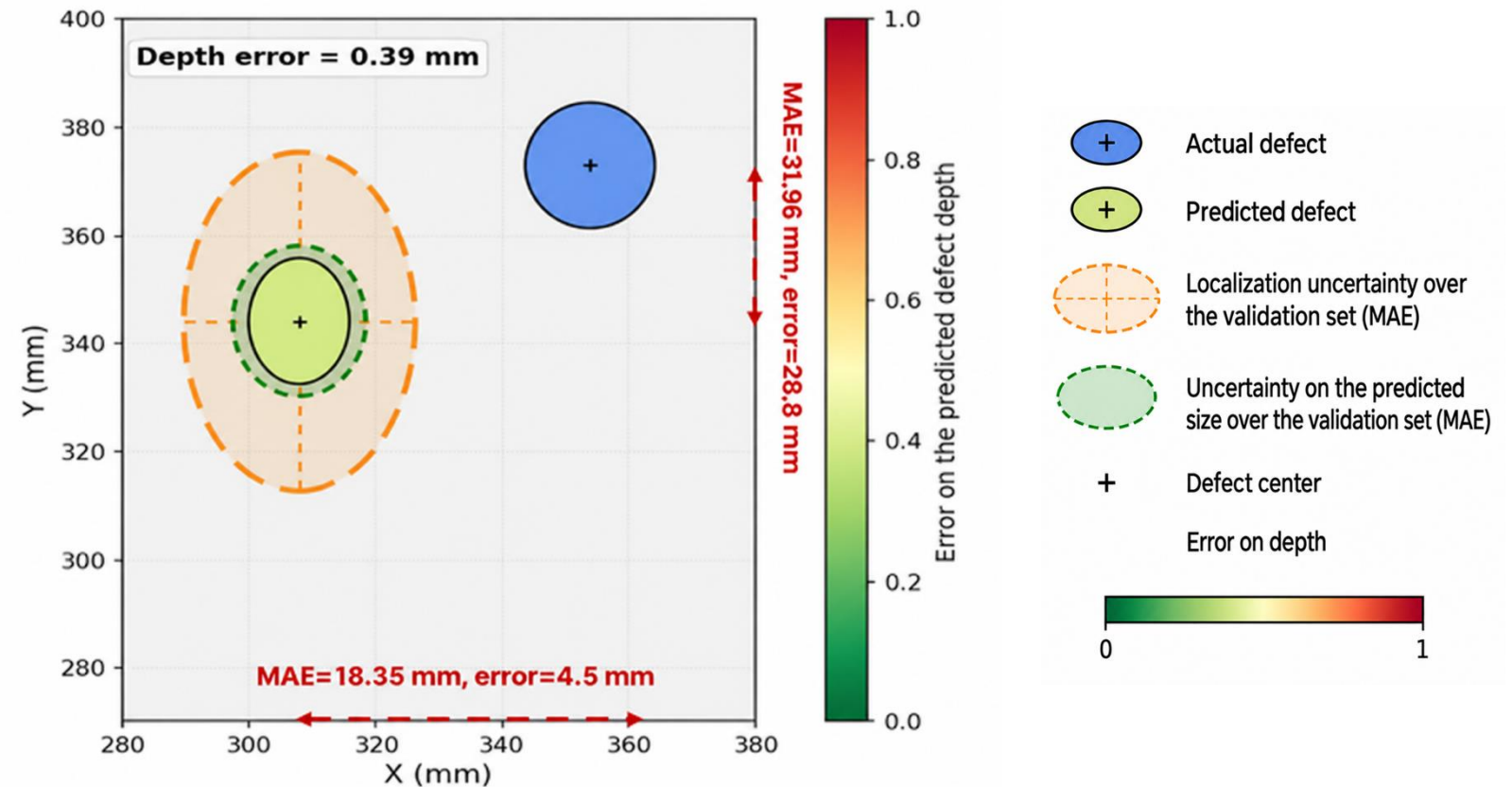
Metric	R^2		
Parameters	Hole Radius	Pos Y	Pos X
Train	0.84	0.98	0.98
Val	0.78	0.97	0.97



Visualization of the actual defect, the predicted defect, and the associated uncertainties

ResNet18 2D - delamination

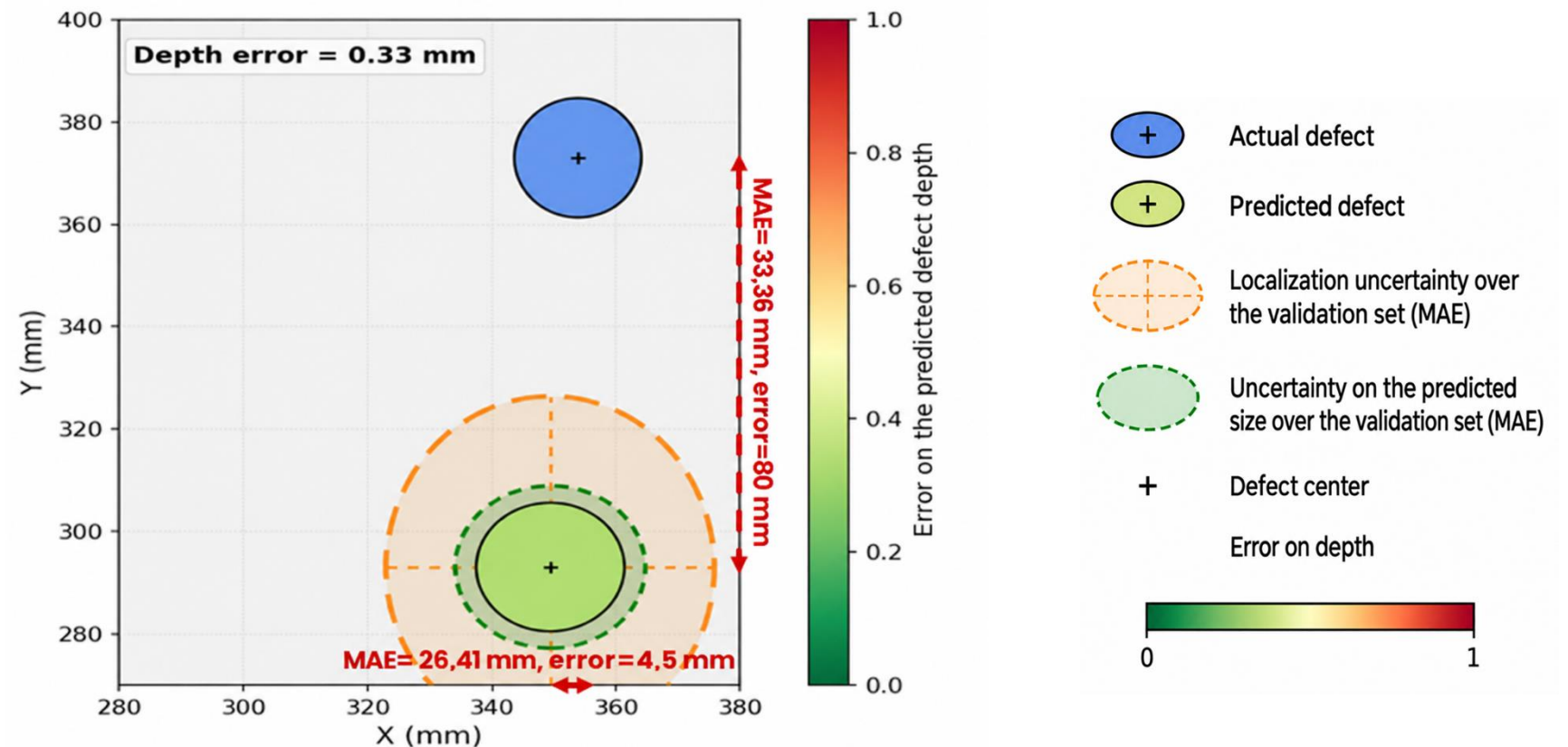
Metric	R^2				
Parameters	Depth	Rx	Ry	Pos Y	Pos X
Train	0.95	0.97	0.98	0.92	0.99
Val	-0.18	0.44	0.47	0.62	0.7



Visualization of the actual defect, the predicted defect, and the associated uncertainties

MobileNet V3 Small- delamination

Metric	R^2				
Parameters	Depth	Rx	Ry	Pos Y	Pos X
Train	0.95	0.93	0.97	0.98	0.97
Val	-0.10	0.13	0.1	0.54	0.51



Visualization of the actual defect, the predicted defect, and the associated uncertainties



Observation:

- **For a hole-type defect:** A clear improvement compared with using raw 1D signals.
- **For a delamination-type defect:** No significant improvement.

Conclusion: Using reference signals improves some performance metrics for hole-type defects, but delamination defects remain very complex and difficult to predict.

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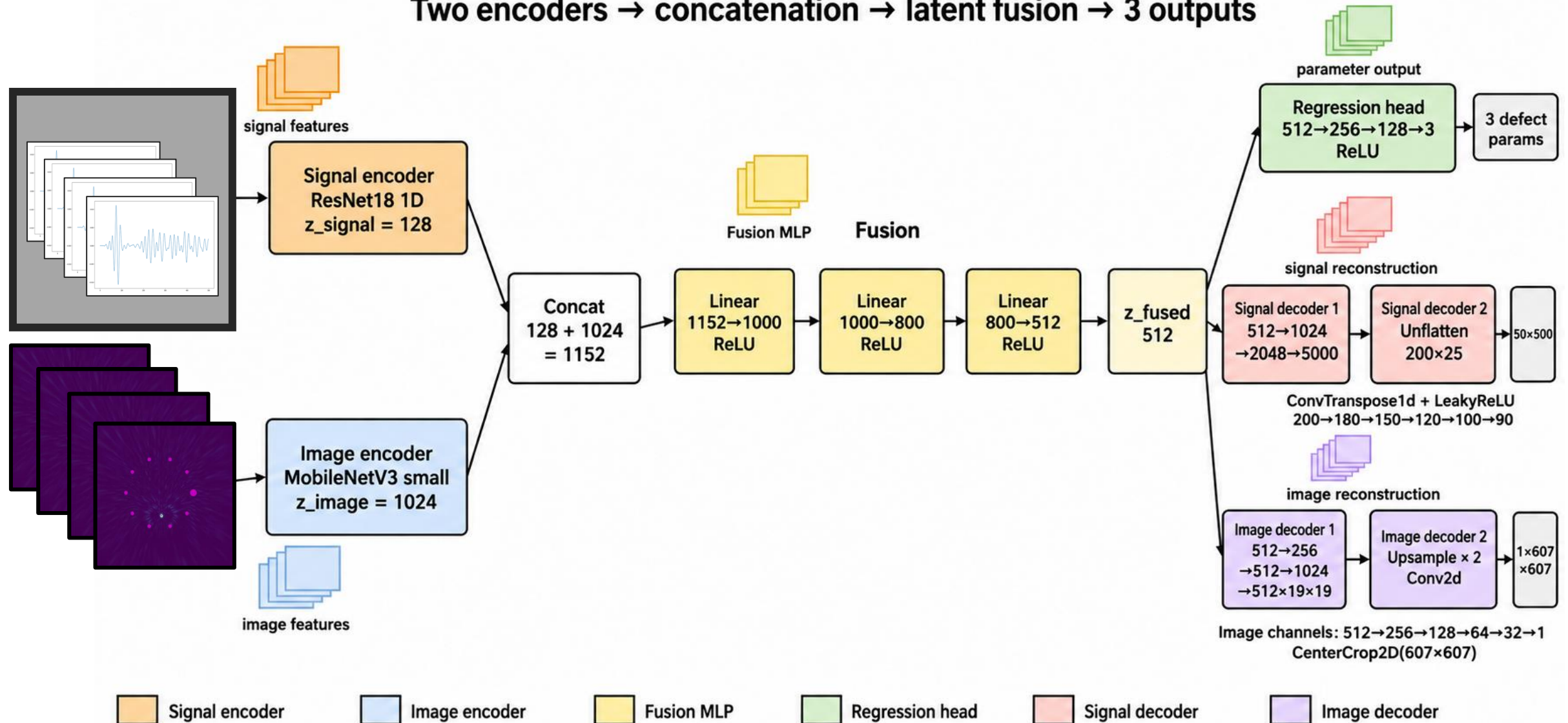
4. Multi-Head Autoencoder

5. Conclusion

To what extent can combining the two modalities reduce the previously observed error, while still allowing partial use of a reference signal?

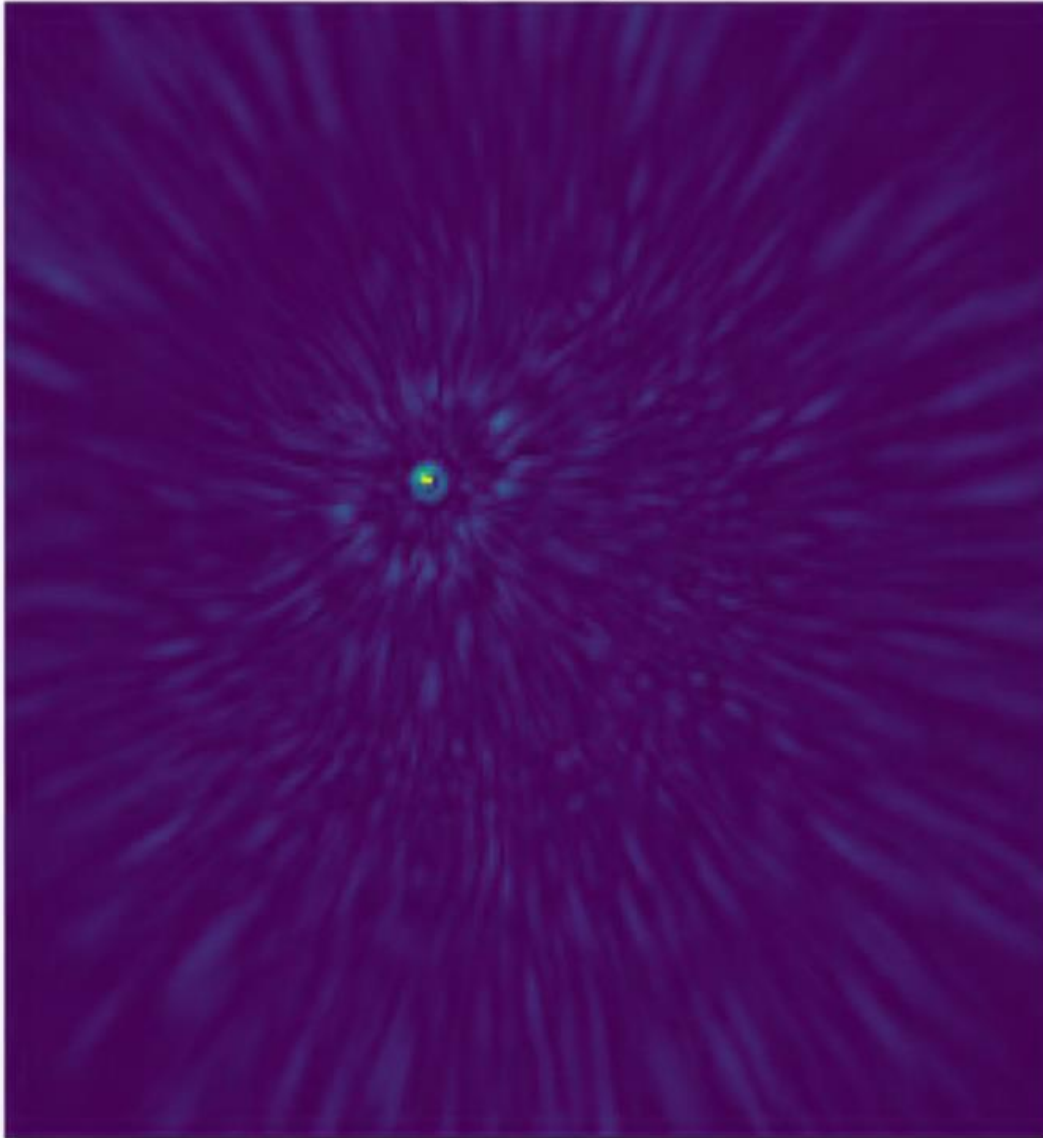
MultiHeadAutoencoder model diagram

Two encoders → concatenation → latent fusion → 3 outputs

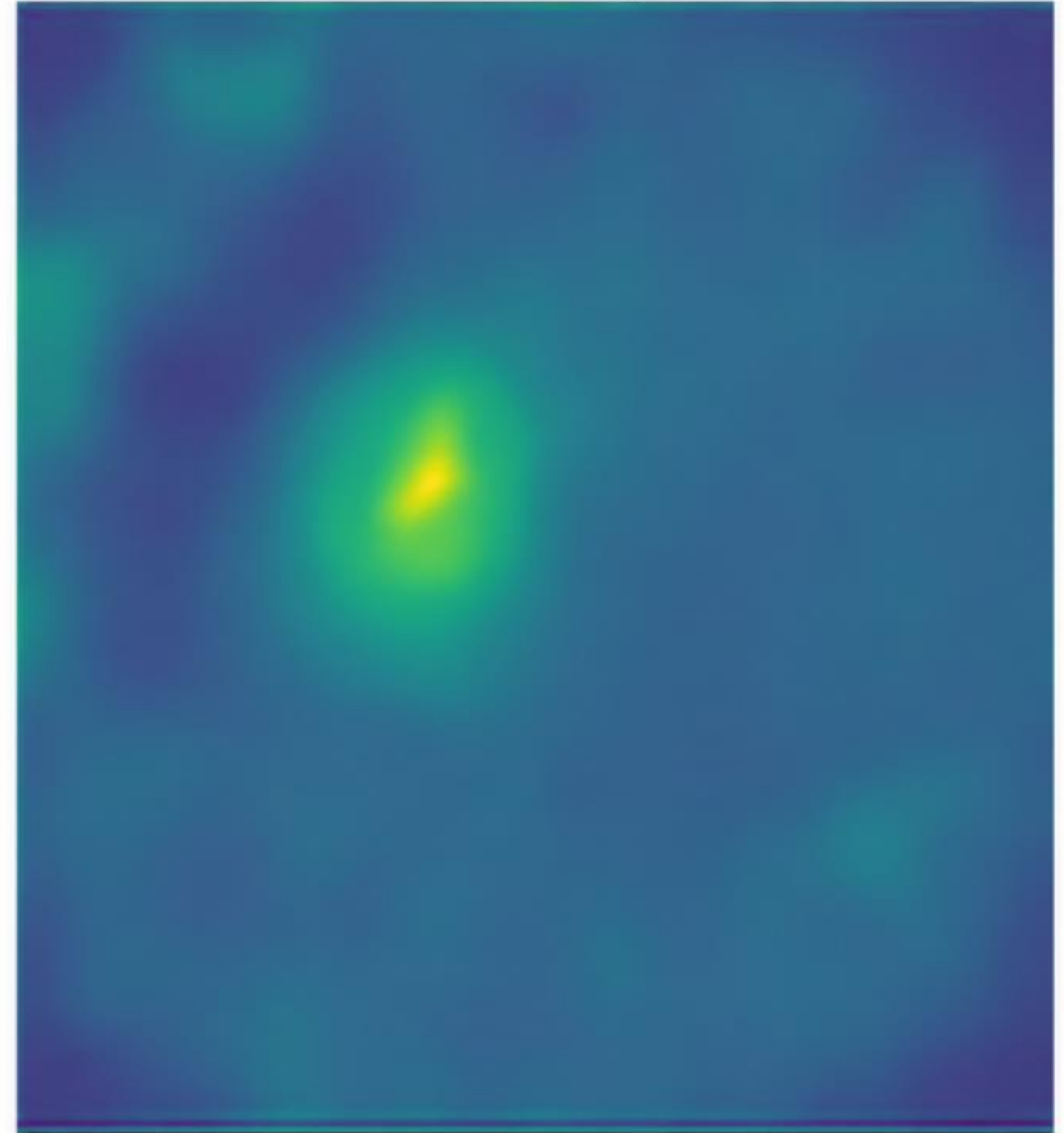




Original image



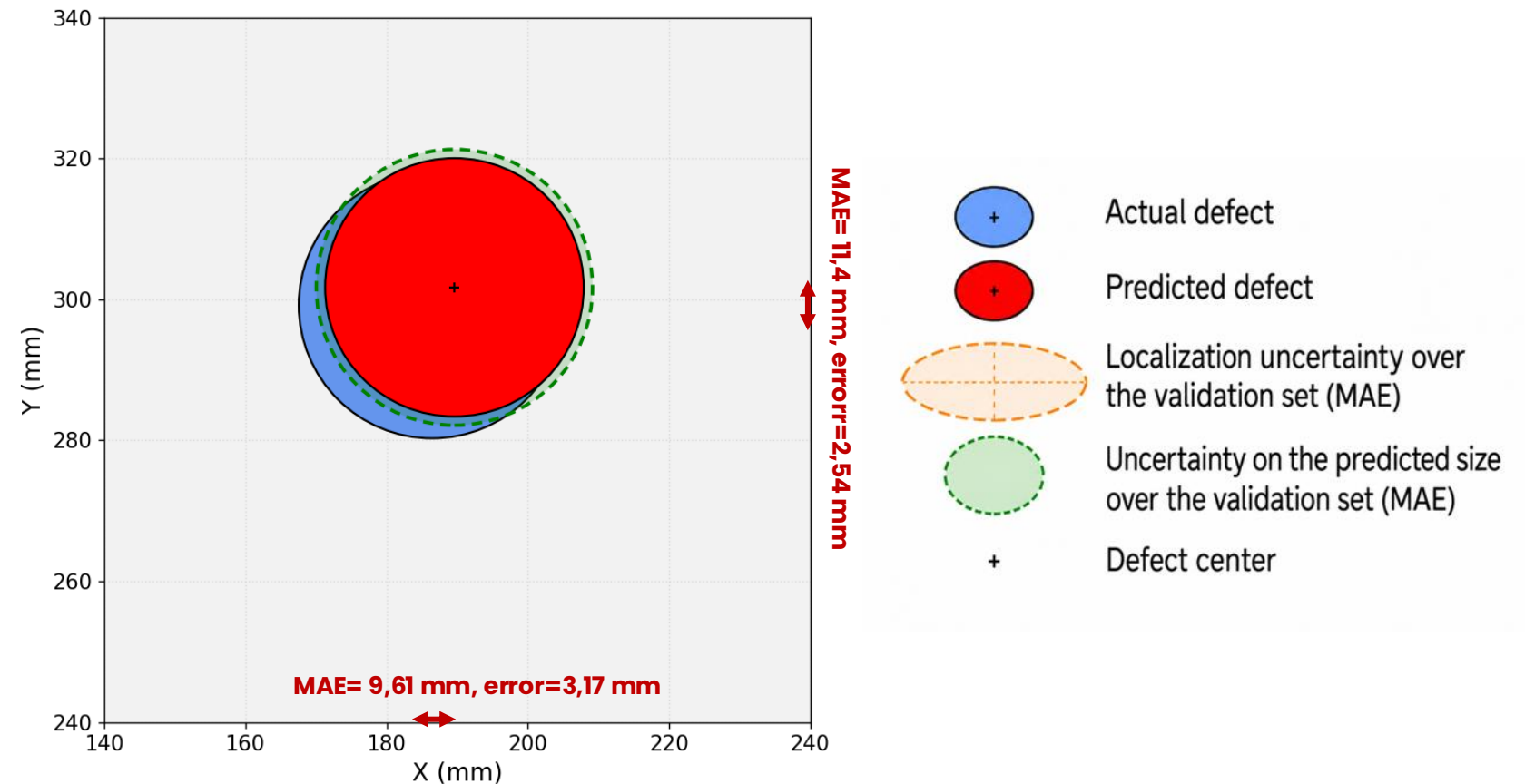
Reconstructed image - both



MultiHead AE – Hole – Modality Dropout

Signals + images

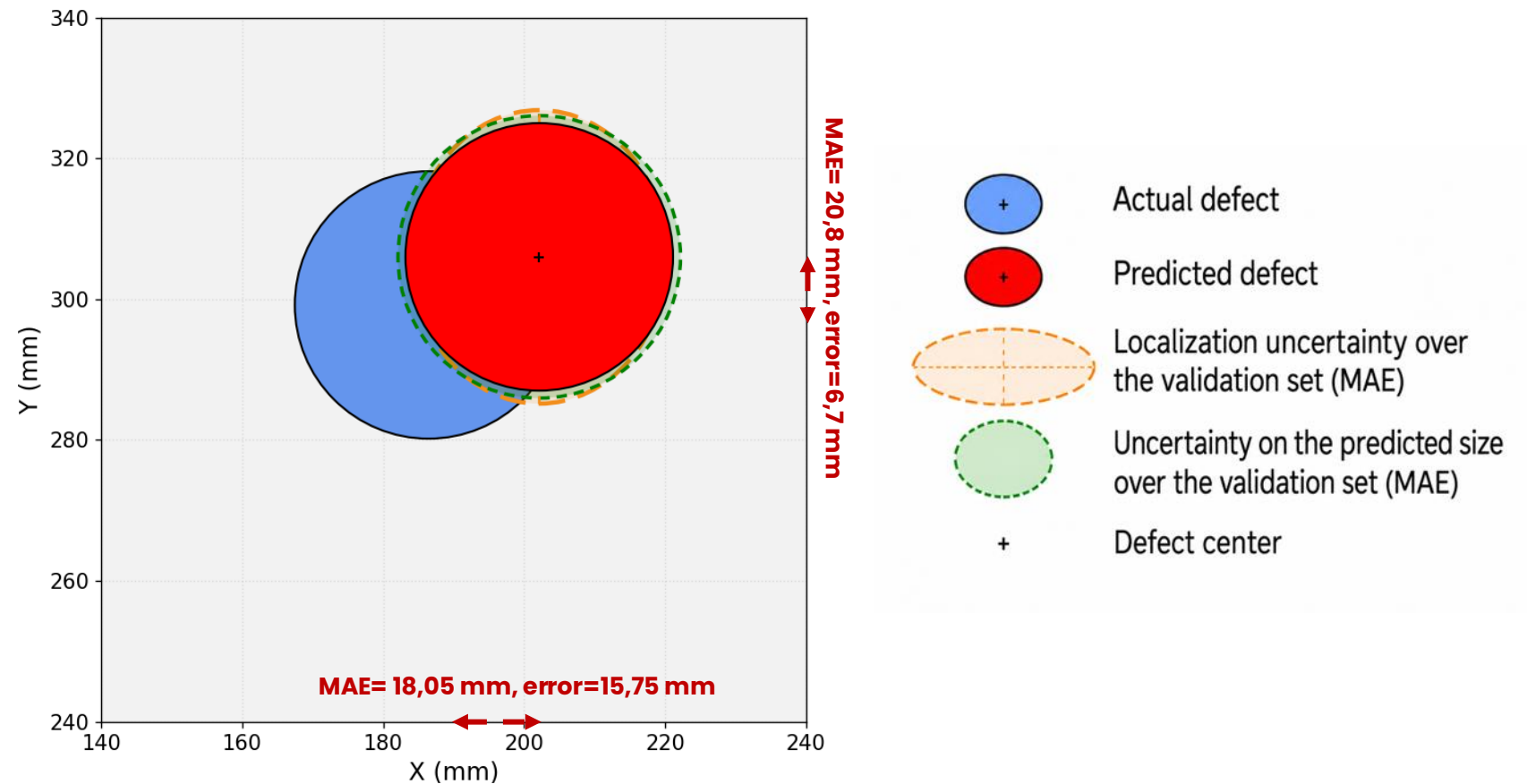
Metric	R^2		
Parameters	Hole Radius	Pos Y	Pos X
Train	0.99	0.96	0.97
Val	0.96	0.96	0.96



Visualization of the actual defect, the predicted defect, and the associated uncertainties

MultiHead AE – Hole – Modality Dropout Signals

Metric	R^2		
Parameters	Hole Radius	Pos Y	Pos X
Train	0.93	0.89	0.89
Val	0.9	0.86	0.83

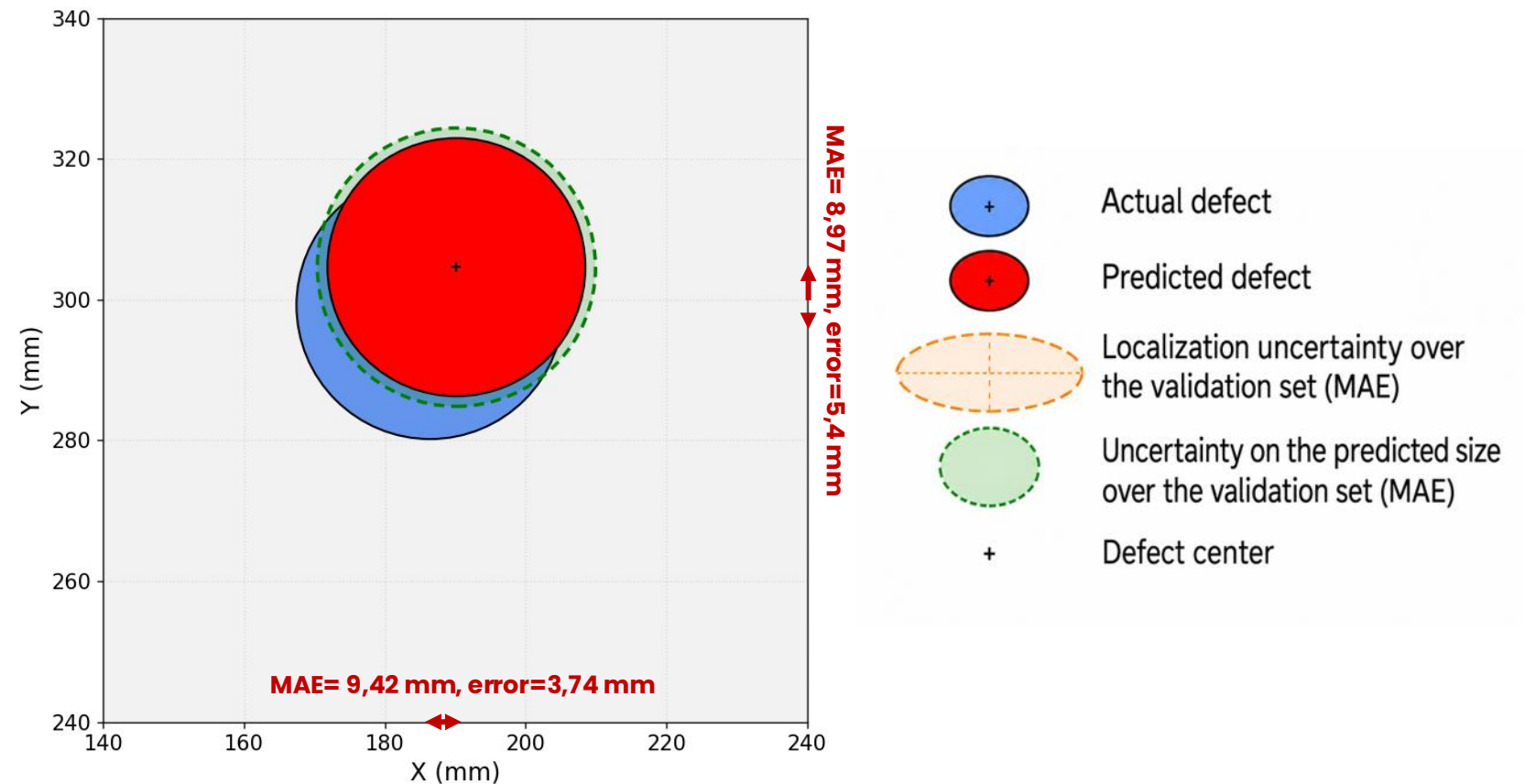


Visualization of the actual defect, the predicted defect, and the associated uncertainties

MultiHead AE – Hole – Modality Dropout

Images

Metric	R^2		
Parameters	Hole Radius	Pos Y	Pos X
Train	0.99	0.98	0.96
Val	0.95	0.96	0.97



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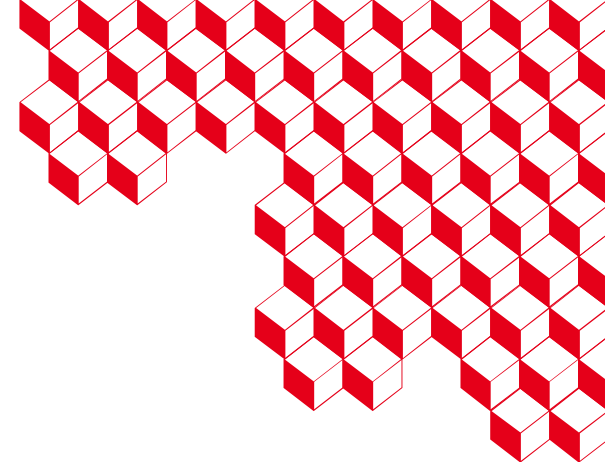
4. Multi-Head Autoencoder

5. Conclusion



- Promising results with state-of-the-art 1D and 2D convolutional neural networks.
1. For the autoencoder, whether the input consists of signals + images or images only, the localization error remains on the order of $\lambda/2$.
 2. For AmiraNet, the localization error is lower, ranging between 15 and 17 mm.

- Lack of experimental validation
- Next goal: use of the composite-material dataset



Thank you for your attention.